

Compte rendu:

Déploiement d'une Architecture Web avec Apache2, DNS et Reverse Proxy Nginx

Hosni Zaaraoui

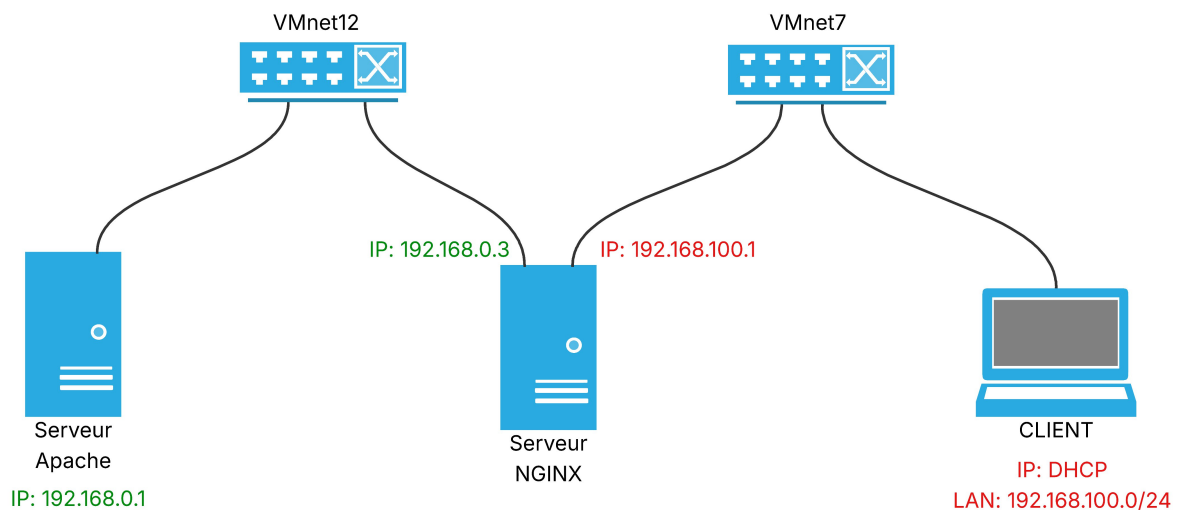
ASSCC1.2

Objectif

Ce CTP a pour objectif de mettre en place une **infrastructure web complète** combinant :

1. Un **serveur Apache2** hébergeant plusieurs sites web sur différents ports.
2. Un **service DNS local** permettant la résolution des noms de domaines internes.
3. Un **serveur Nginx** configuré comme **reverse proxy**, servant d'interface d'accès unique aux différents sites hébergés sur Apache.

Cette architecture illustre une approche moderne de la gestion des services web, en séparant les rôles entre la couche applicative et la couche d'accès.



1. Installation et vérification du serveur Apache2

Je commence par installer Apache2 :

```
sudo apt-get install apache2
```

```
hosniadmin@SERVER-ALPHA: ~/Desktop
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0
0 upgraded, 9 newly installed, 0 to remove and 380 not upgraded.
Need to get 2,063 kB of archives.
After this operation, 8,238 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Ign:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64
  1.7.0-8ubuntu0.22.04.2
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 a
md64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-d
```

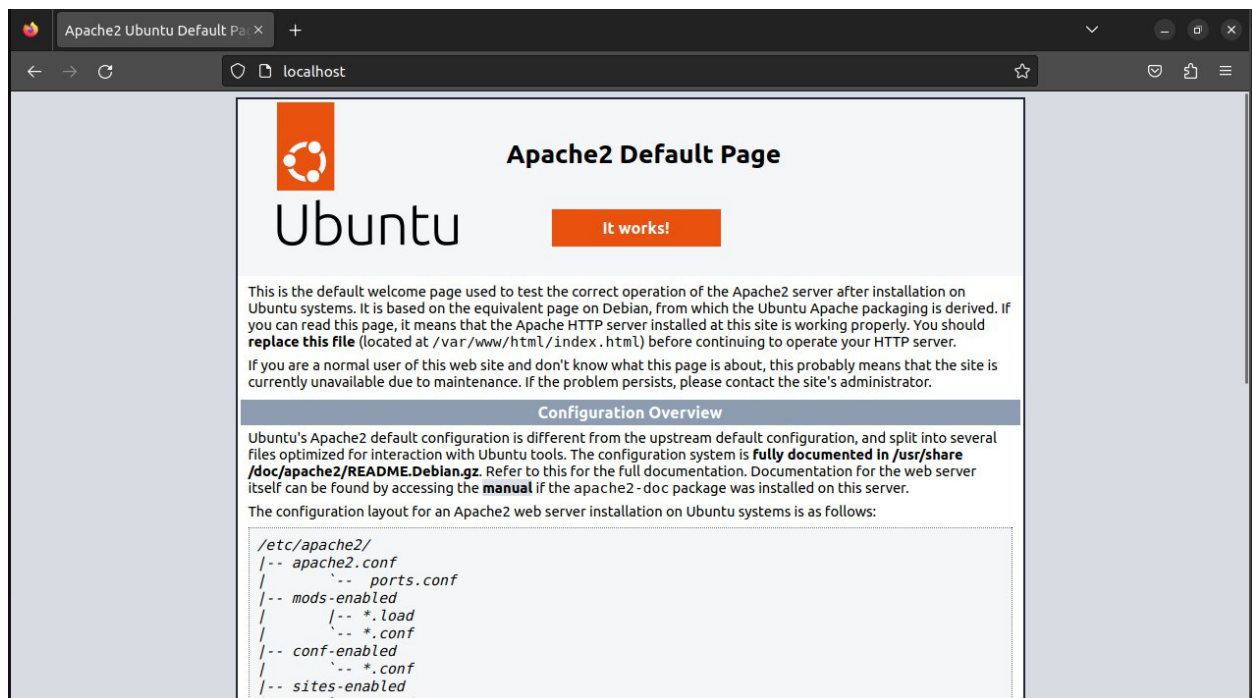
Ensuite, je vérifie que le serveur fonctionne correctement :

```
wget 127.0.0.1
```

```
hosniadmin@SERVER-ALPHA: ~/Desktop
hosniadmin@SERVER-ALPHA:~/Desktop$ wget 127.0.0.1
--2025-11-19 09:21:30--  http://127.0.0.1/
Connecting to 127.0.0.1:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10671 (10K) [text/html]
Saving to: 'index.html'

index.html      100%[=====>]  10.42K  --.-KB/s    in 0s
2025-11-19 09:21:30 (58.0 MB/s) - 'index.html' saved [10671/10671]
```

et en accédant à <http://localhost> depuis un navigateur.



Le serveur répond sur le port 80, confirmant le bon fonctionnement d'Apache.

2. Activation du SSL et génération du certificat

Pour sécuriser le site principal, j'active le module SSL :

```
sudo a2enmod ssl
```

```
hosniadmin@SERVER-ALPHA:/etc/apache2/ssl$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
systemctl restart apache2
```

```
sudo systemctl restart apache2
```

Je crée ensuite le répertoire des certificats :

```
sudo mkdir /etc/apache2/ssl
```

et je génère un certificat auto-signé :

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/apache2/ssl/apache.key \
-out /etc/apache2/ssl/apache.crt
```

```
.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....  
.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....  
.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....  
+++  
-----  
You are about to be asked to enter information that will be incorporated  
into your certificate request.  
What you are about to enter is what is called a Distinguished Name or a DN.  
There are quite a few fields but you can leave some blank  
For some fields there will be a default value,  
If you enter '.', the field will be left blank.  
-----  
Country Name (2 letter code) [AU]:tn  
State or Province Name (full name) [Some-State]:Rt mahdia KM2  
Locality Name (eg, city) []:sfax  
Organization Name (eg, company) [Internet Widgits Pty Ltd]:ISET SFAX  
Organizational Unit Name (eg, section) []:Departement Informatique  
Common Name (e.g. server FQDN or YOUR name) []:www.isetsf.rnu.tn  
Email Address []:hosnyzaaraoui@gmail.com  
  
Please enter the following 'extra' attributes  
to be sent with your certificate request
```

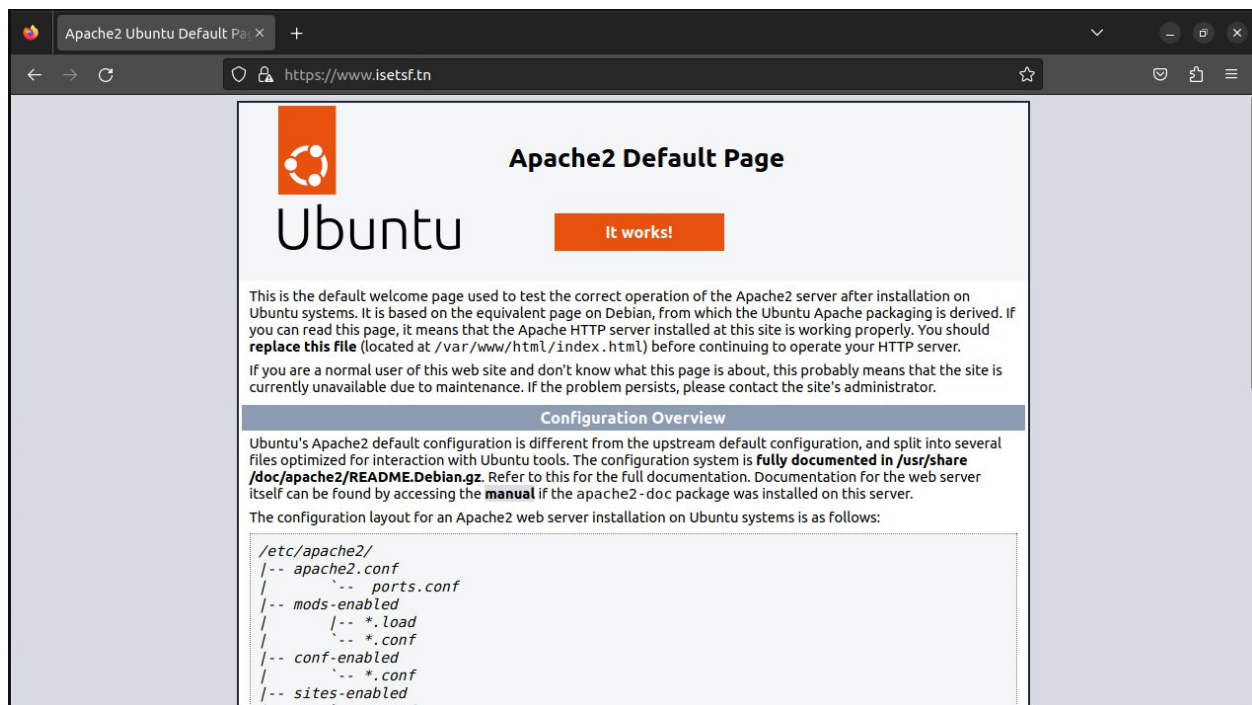
Je configure le fichier `default-ssl.conf` pour utiliser ces certificats, puis j'active le site :

```
sudo a2ensite default-ssl.conf
```

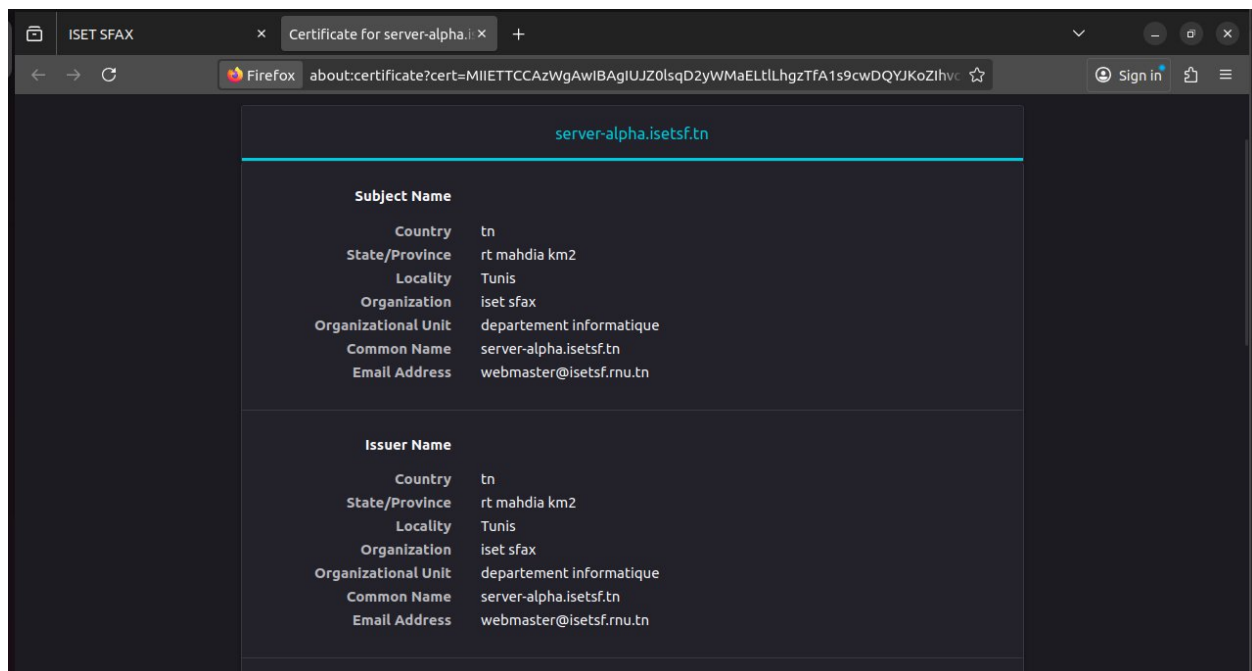
```
sudo service apache2 reload
```

```
hosniadmin@SERVER-ALPHA:/etc/apache2/sites$ cd ../sites-available/
hosniadmin@SERVER-ALPHA:/etc/apache2/sites-available$ sudo a2ensite default-ssl.conf
Enabling site default-ssl.
To activate the new configuration, you need to run:
    systemctl reload apache2
hosniadmin@SERVER-ALPHA:/etc/apache2/sites-available$
```

Je teste la connexion en accédant à <https://www.isetsf.tn>.



Le site s'affiche correctement avec un certificat SSL valide (auto-signé).



3. Configuration du DNS local

Je configure le service DNS Bind9 (hébergé sur le même serveur qu'Apache) afin de gérer les domaines internes :

- `www.asscc1.net` → **adresse du serveur Nginx**,
- `www.asscc2.org` → **adresse du serveur Nginx**.

Chaque zone contient un enregistrement A pointant vers 192.168.0.3, l'adresse du futur serveur Nginx.

Je crée les fichiers de zones `db.asscc1.net` et `db.asscc2.org`, puis je les déclare dans le fichier `named.conf.local`.

Cela permet de rediriger automatiquement le trafic HTTP vers le proxy inverse.

```
hosniadmin@SERVER-ALPHA: /etc/apache2/sites-available
GNU nano 6.2 /etc/bind/db.asscc1.net
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA server-alpha.isetsf.tn. hosnyzaaraoui.gmail.com. (
    20251911 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS server-alpha.isetsf.tn.
server-alpha IN A 192.168.0.1
www IN A 192.168.0.3
```

```
GNU nano 6.2 named.conf.local
zone "isetsf.tn" {
    type master;
    file "/etc/bind/db.isetsf.tn";
};

zone "asscc1.net" {
    type master;
    file "/etc/bind/db.asscc1.net";
};

zone "asscc2.org" {
    type master;
    file "/etc/bind/db.asscc2.org";
};

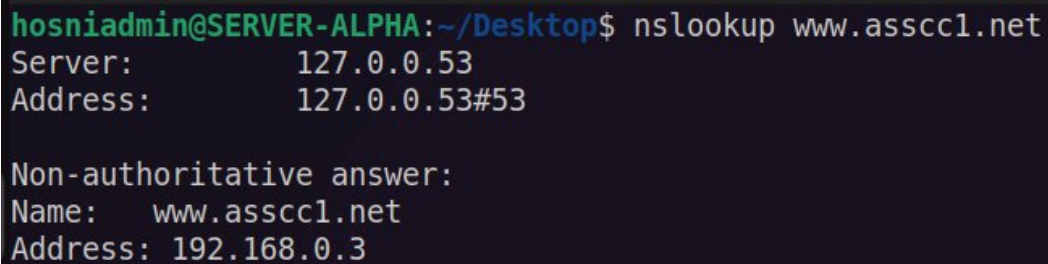
zone "0.168.192.in-addr.arpa" {
    type master;
    notify no;
    file "/etc/bind/db.192";
};
```

Après la configuration, je redémarre le service :

```
sudo service bind9 restart
```

et je teste :

```
nslookup www.asscc1.net
```

A terminal window with a dark background. The prompt is 'hosniadmin@SERVER-ALPHA:~/Desktop\$'. The command 'nslookup www.asscc1.net' has been executed. The output shows the server IP as 127.0.0.53 and the address as 127.0.0.53#53. Below this, it indicates a 'Non-authoritative answer' with the name 'www.asscc1.net' and the address '192.168.0.3'.

```
hosniadmin@SERVER-ALPHA:~/Desktop$ nslookup www.asscc1.net
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   www.asscc1.net
Address: 192.168.0.3
```

Les réponses confirment la bonne résolution vers le serveur Nginx.

4. Hébergement multi-sites Apache

Je configure maintenant Apache pour héberger plusieurs sites distincts.

Je crée deux nouveaux répertoires :

```
sudo mkdir -p /var/www/html/asscc1
```

```
sudo mkdir -p /var/www/html/asscc2
```

et j'y place un fichier index.html différent pour chaque site.

Je crée ensuite les fichiers de configuration :

- /etc/apache2/sites-available/asscc1.conf pour le site www.asscc1.net (port 8001),
- /etc/apache2/sites-available/asscc2.conf pour le site www.asscc2.org (port 8002).

```
hosniadmin@SERVER-ALPHA: /etc/apache2/sites-available
GNU nano 6.2 asscc1.conf
<VirtualHost *:8001>

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/asscc1
    ServerName www.asscc1.net
    ServerAlias asscc1.net

    ErrorLog ${APACHE_LOG_DIR}/asscc1_error.log
    CustomLog ${APACHE_LOG_DIR}/asscc1_access.log combined

</VirtualHost>
```

J'ajoute les ports dans ports.conf :

```
hosniadmin@SERVER-ALPHA: /etc/apache2/sites-available
GNU nano 6.2 ../ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 80
Listen 8001
Listen 8002

<IfModule ssl_module>
    Listen 443
</IfModule>
```

et j'active les nouveaux sites :

```
sudo a2ensite asscc1.conf
```

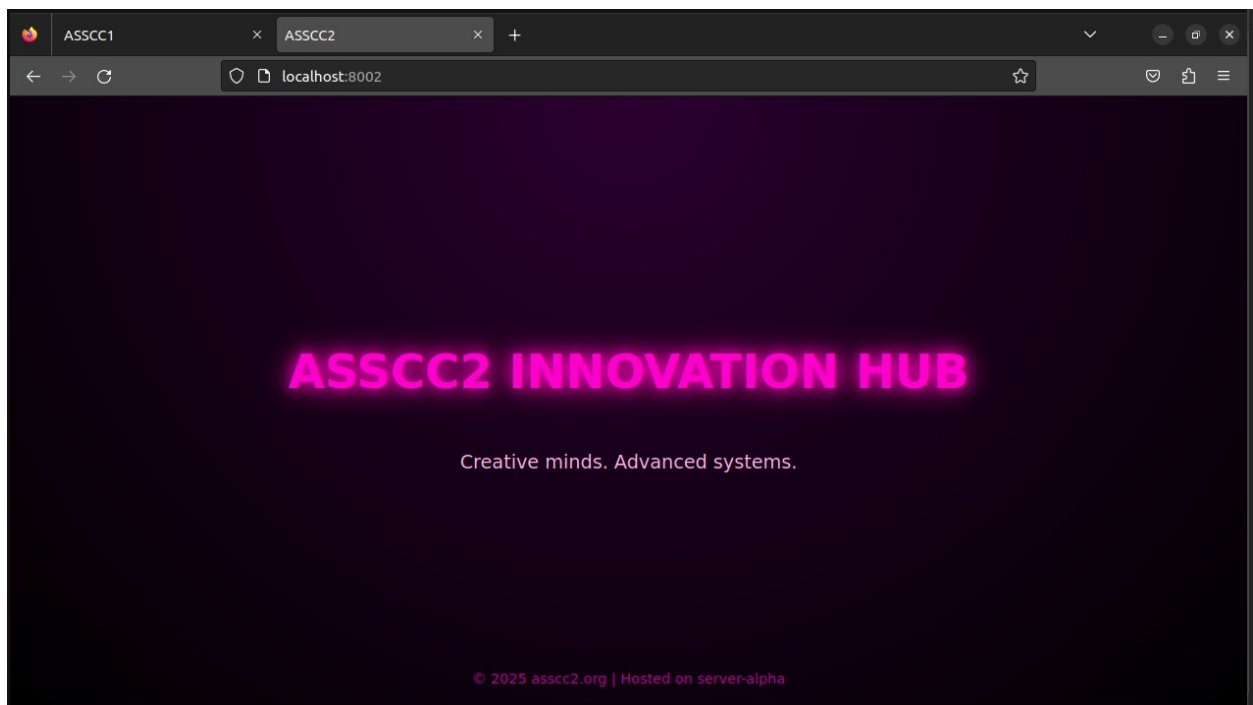
```
sudo a2ensite asscc2.conf
```

```
sudo service apache2 restart
```

Je teste ensuite chaque site depuis le navigateur :

<http://localhost:8001>

<http://localhost:8002>



Les deux pages s'affichent correctement, ce qui prouve le bon fonctionnement de la configuration multi-sites.

5. Mise en place du Reverse Proxy Nginx

Sur une seconde machine virtuelle, j'installe le serveur Nginx :

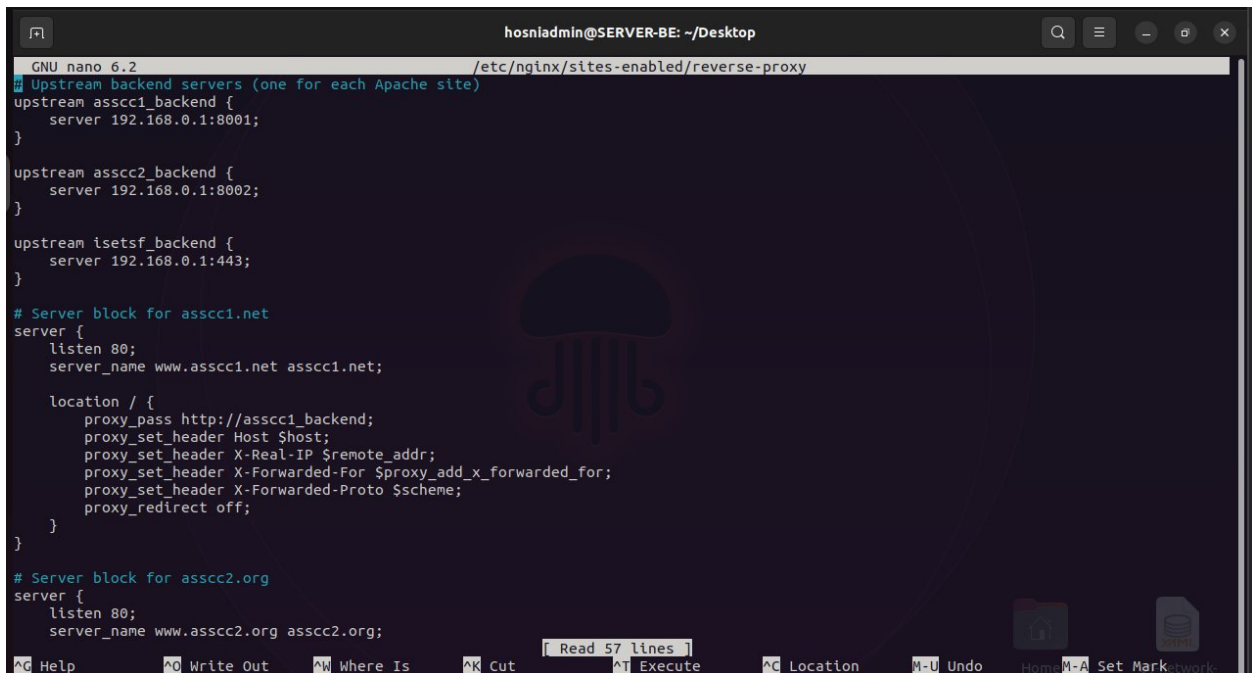
```
sudo apt install nginx
```

```
hosniadmin@SERVER-BE:~/Desktop$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.18.0-6ubuntu14.7).
0 upgraded, 0 newly installed, 0 to remove and 441 not upgraded.
hosniadmin@SERVER-BE:~/Desktop$
```

Je configure Nginx comme reverse proxy afin qu'il redirige les requêtes externes vers le serveur Apache interne selon le domaine demandé.

Le fichier de configuration contient les redirections suivantes :

- www.isetsf.tn → Apache (HTTPS, port 443)
- www.asscc1.net → Apache (port 8001)
- www.asscc2.org → Apache (port 8002)



```
GNU nano 6.2 /etc/nginx/sites-enabled/reverse-proxy
# Upstream backend servers (one for each Apache site)
upstream asscc1_backend {
    server 192.168.0.1:8001;
}

upstream asscc2_backend {
    server 192.168.0.1:8002;
}

upstream isetsf_backend {
    server 192.168.0.1:443;
}

# Server block for asscc1.net
server {
    listen 80;
    server_name www.asscc1.net asscc1.net;

    location / {
        proxy_pass http://asscc1_backend;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_redirect off;
    }
}

# Server block for asscc2.org
server {
    listen 80;
    server_name www.asscc2.org asscc2.org;
```

Après avoir redémarré Nginx, je teste depuis un client.

Les trois sites sont accessibles à partir de leur nom de domaine respectif sans indiquer le port, preuve que le reverse proxy fonctionne correctement.

ASSCC1 Network Portal

Empowering connectivity through innovation.

200 GET www.asscc1.net / document 785 B 74...

2 requests 1.03 kB / 785 B transferred Finish: 279 ms DOMContentLoaded: 221

Response Headers (304 B)

- Accept-Ranges: bytes
- Connection: keep-alive
- Content-Encoding: gzip
- Content-Length: 481
- Content-Type: text/html
- Date: Wed, 19 Nov 2025 19:26:25 GMT
- ETag: "2ed-643f6e8eb1440-gzip"
- Last-Modified: Wed, 19 Nov 2025 18:40:09 GMT
- Server: nginx/1.18.0 (Ubuntu)
- Vary: Accept-Encoding

ASSCC2 INNOVATION HUB

Creative minds. Advanced systems.

© 2025 asscc2.org | Hosted on server-alpha

200 GET www.asscc2.org / document 842 B 84...

2 requests 1.01 kB / 842 B transferred Finish: 760 ms DOMContentLoaded: 620

Response Headers (304 B)

- Accept-Ranges: bytes
- Connection: keep-alive
- Content-Encoding: gzip
- Content-Length: 538
- Content-Type: text/html
- Date: Wed, 19 Nov 2025 19:27:01 GMT
- ETag: "34e-643f6ea3ac5d0-gzip"
- Last-Modified: Wed, 19 Nov 2025 18:40:31 GMT
- Server: nginx/1.18.0 (Ubuntu)
- Vary: Accept-Encoding

6. Tests et validation

Je teste la résolution DNS, la redirection du proxy et la réponse HTTP/HTTPS de chaque site :

- Les noms de domaine sont correctement résolus.

```
C:\Users\Hosni>nslookup www.asscc1.net
Server: www.isetsf.tn
Address: 192.168.0.1

Name: www.asscc1.net
Address: 192.168.0.3

C:\Users\Hosni>nslookup www.asscc2.org
Server: www.isetsf.tn
Address: 192.168.0.1

Name: www.asscc2.org
Address: 192.168.0.3
```

- Les sites répondent via Nginx.
- Le certificat SSL s'affiche pour www.isetsf.tn.

